

Identification of metabolites in human plasma and urine after consumption of a polyphenol-rich juice drink.

A polyphenol-rich (P-R) juice drink was developed as a potential approach to increase intake of dietary polyphenols. Analysis of the beverage by HPLC with PDA, fluorescence, and MS detection facilitated the identification/partial identification of 40 flavonoids and related phenolic compounds. The main constituents were (-)-epigallocatechin and other green tea flavan-3-ols, phloretin-2'-O-glucoside, gallic acid, hesperetin-7-O-rutinoside, 5-O-caffeoylquinic acid, and procyanidins, with trace levels of several flavonols and purple grape juice anthocyanins also being present. Healthy human subjects (n = 10) consumed 350 mL of the P-R juice drink, after which plasma and urine samples were collected over a 0-24 h period. HPLC-MS analysis identified 13 metabolites in plasma and a further 20 in urine. Qualitatively, the profiles of the glucuronide, sulfated, and methylated metabolites were very similar to those detected in earlier investigations when the main components in the juice drink were consumed separately in feeding studies with coffee, green tea, orange juice, and apple cider.